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From thriving to task focus: the role of needs-supplies fit and task complexity

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Abstract

Can thriving at work be a self-sustaining phenomenon? In our study we incorporated the person-environment fit perspective into socially embedded model of thriving by Spreitzer's et al. (*Organization Science*, *16*(5), 537–549, 2005) to explore how and when thriving affects individuals' task focus, an agentic behavior often considered an antecedent of thriving. We proposed that individuals who thrive at work tend to perceive that the rewards supplied by the job meet their needs of growth and development, which leads to more focus on their tasks. We also proposed that task complexity interacts with thriving to influence the mechanism of needs-supplies (N-S) fit. Based on two-wave data from 170 product engineers, the results of our study showed that thriving indirectly affects task focus through perceived N-S fit. When faced with high-complexity tasks, high-thriving employees generated higher N-S fit perceptions. When thriving was low, N-S fit was highest in in the low-complexity task context, suggesting that matching thriving to task complexity could be an important strategy by which managers might maintain higher levels of task focus, which we speculate could in turn promote thriving. Our study advances research on thriving, fit perceptions and work behavior by revealing these relationships.

Keywords Thriving at work · Task focus · N-S fit · Task complexity

Introduction

In the past decade, thriving at work (TAW), a psychological state in which people experience both learning and vitality (Spreitzer et al., 2005), has received considerable interest in the growing movement of positive organizational scholarship (Goh et al., 2022; Rizqi et al., 2023). This satisfying subjective experience is strongly associated with a series of important outcomes, with thriving employees found to be healthier, more productive, engaged, and creative (e.g. Jiang et al., 2021; Wallace et al., 2016). According to the socially embedded model of thriving (SEMT) proposed by Spreitzer et al. (2005), there is a feedback loop between thriving at work and agentic work behavior, through which thriving may become self-sustaining. However, although some research supports the proposed positive effects of agentic work behavior on thriving, there is insufficient empirical evidence on whether and how thriving leads to such agentic work behavior, such as task focus (Niessen et al., 2012). Establishing a link to an antecedent of thriving is particularly important because, although thriving employees may reinvest their energies directly into work, they may seek alternative opportunities for self-development such as taking up part-time jobs or cultivating hobbies (Hyde et al., 2022; Porath et al., 2022; Chew, 2017). Indeed, a diary study conducted by Niessen et al. (2012) found that thriving on a prior day did not significantly predict task focus on the subsequent day. This finding is also consistent with observations in the organizational behavior literature that positive states do not necessarily improve task performance in all contexts (e.g. Ashford et al., 2018). Further exploration of thriving as an antecedent of task focus will add to our

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understanding of the conditions under which the reciprocal relationship can take hold.

We propose that the mechanism by which thriving influences agentic work behavior is needs-supplies (N-S) fit, an assessment of the extent to which the workplace supplies the appropriate environment, means, outcomes, or rewards to satisfy the individual's psychological needs, values, and goals (Kristof-Brown, 1996). N-S fit is the congruence between individual preferences for these characteristics and the characteristics actually present in the work (Shaw & Gupat, 2004), forming an important link between subjective states and behavior (Yu, 2009). Individuals may use their feeling of thriving to inform their judgement and evaluation of whether their present job or task provides for their needs. Specifically, we propose that individuals infer the extent to which their work supplies (S) meet their needs (N) from their level of energy, growth, and progress. Then, N-S fit signals the intrinsic rewards of the task while clearing the mind of the frustration and distraction of a needs-supplies misalignment, thereby motivating task related agentic behavior, which we have operationalized as task focus.

In order to achieve sustainable thriving, thriving individuals may prefer increasingly challenging tasks and develop their capacity to meet these challenges (Spreitzer et al., 2012). Consistent with principles of person-environment fit, whereby people tend to select and engage in work environments that are compatible with their skills, interests, values, and other personal characteristics (Kristof-Brown et al., 2005; Milliman et al., 2017), task complexity can act as a moderator, with high- thriving individuals assessing greater N-S fit when task complexity is correspondingly high. Our full moderated mediation model is presented in Fig. 1.

In summary, by incorporating a needs-supplies fit perspective into the SEMT, the aim of this study is to explore how and when thriving influences employees' focus on tasks. this study contributes to existing literatures in three ways. First, we contribute to the nascent literature on the psychological mechanisms through which thriving affects individual's behavior. We test the central route from thriving to task focus by examining perceived N-S fit as a mediator, which expands our understanding of the socially embedded model of thriving. Second, we offer a first empirical investigation of task complexity as an important boundary condition for the effect of thriving on a needs-supplies fit judgment, hence adding to a more diversified picture of the task-context's role in shaping thriving and its outcomes. Third, our study promotes convergence and understanding of the two research fields of thriving and fit.

Theoretical framework and hypotheses

Thriving and task focus

Thriving consists of two essential components: vitality and learning. Vitality refers to the positive feeling of having energy, while learning is more of a cognitive experience, a sense that an individual is acquiring and can apply knowledge and skills (Spreitzer et al., 2005). When vitality and learning are simultaneously at high levels, employees achieve a state of thriving at work.

Individuals are more likely to focus on the tasks at hand when they experience thriving. First, the vitality dimension of thriving at work provides the energy to maintain the necessary focus and alertness while performing tasks (Porath et al., 2012). Schaufeli and Bakker (2004) found that the thriving individual tends to exhibit a high level of energy and perseverance as well as the capacity to work voluntarily and consciously. In addition, the broaden-and-build theory of positive emotions (Fredrickson, 2001) suggests that positive affect associated with thriving can expand an individual's range of attention, cognition, and action. For example, Erez and Isen (2002) found that subjects with positive moods stayed in their experiment longer than those with neutral moods. A study by Martin et al. (1993) of a community sample of 208 adults showed that happy participants had higher levels of focus and were able to stay on task longer than those who were unhappy or depressed. Third, vitality and learning can increase one's beliefs in own capabilities and optimism in meeting their goals, which lead

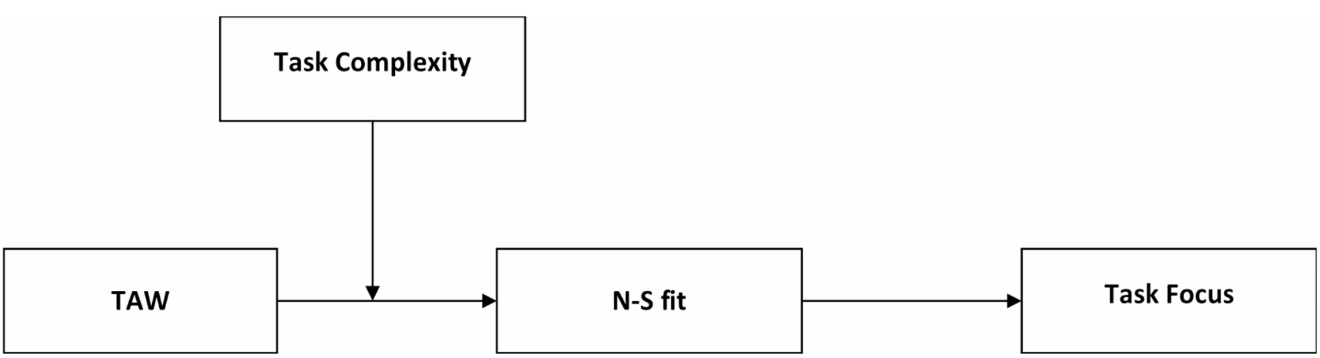

Fig. 1 Moderated mediation model for the relationship between thriving at work and task focus

to higher levels of work engagement (Xanthopoulou et al., 2009). When people experience growth in their work, they tend to gain self-efficacy (Bensemmane et al., 2018) and be more confident about mastering job requirements (Sonenshein et al., 2013). For example, an experimental study by Tasi et al. (2007) found that participants with positive moods generated higher self-efficacy, which led to greater task persistence.

Based on these rationales and evidence, we hypothesize that:

Hypothesis 1: Thriving at work is positively related to employees' work focus.

The mediating role of N-S fit

N-S fit is the agreement between an individual's needs, defined as preferences, interests, motives, and goals, and the environmental supplies, which at work can be monetary rewards, benefits, and job characteristics (Kristof-Brown et al., 2005). Spreitzer et al. (2005), in their socially embedded theory of thriving, proposed that thriving can help individuals estimate whether what they are doing is beneficial to their development. Thriving implicates needs for learning, growing, and, more broadly, positive subjective well-being (Spreitzer et al., 2005).

We propose that when people experience thriving at work, they are more likely to assess a strong perception of N-S fit. Similar to the feelings-as-information theory, whereby individuals tend to regard their own feelings as the source of information that influences their subsequent judgements (Clore et al., 1994), individuals may use their feeling of thriving as an informational cue by which to gauge whether the supplies of work meet their needs (Yu & David, 2016). The essence of this process is to ask one's self "Is this the job I want?" Individuals who thrive at work are more likely to make a positive judgment that "the job supplies (S) meets my needs (N)." Conversely, when individuals feel they are not thriving at work, they are more likely to develop a negative perception that "my needs (N) are not being met at work (S)." Besides this relatively direct assessment that uses thriving as a proxy for other information, the principle of emotional congruence suggests that positive mood makes positive information more accessible in memory, whereas negative mood similarly makes negative information more accessible (Schwarz & Strack, 1991).

We propose that perceived fit is positively correlated with task focus. The underlying logic of the employment relationship is that people take and stick with a job primarily based on the rewards of the job, and that people invest their time and energy at work to get those rewards. Individuals whose self-needs are met in the organization will experience job meaning and self-worth, and thus become more committed to their work role (Van et al., 2022). Indeed, a meta-analysis conducted by Kristof-Brown et al. (2005) found that N-S fit was strongly associated with job satisfaction, occupational satisfaction, and affective commitment. In contrast, employees who perceive a N-S misfit may feel bored and unenthusiastic at work, ultimately leading to distraction and deviation from job performance and goals. In short, individuals are more willing to focus on the task at hand in a work environment that fits their skills, interests, values, and other personal traits (Milliman et al., 2017). Based on this, we proposed:

Hypothesis 2: Thriving at work is positively related to task focus through N-S fit.

The moderating role of task complexity

Thriving at work relates positivity to needs for self-development (Paterson et al., 2014). When employees realize that they are making progress through learning on the job, they are likely to look for opportunities to gain additional skills to further development (Porath et al., 2022; Kleine et al., 2019). Task complexity may resonate with these needs by challenging the thriving individual.

Task complexity is defined by the professional skills, reasoning and problem-solving resources, and physical and mental demands required to fulfil a task (e.g. Braarud, 2001; Wood, 1986). Campbell (1988) argued that task complexity is related directly to the attributes of the task that increase information load, diversity, or rate of change (Campbell, 1988), requiring individuals not only to fully apply their previous knowledge and skills but also to learn new approaches and develop the ability to adapt quickly. More complex tasks require more resources (Campbell, 1988), but are also a good opportunity for gaining knowledge, enhancing abilities, and increasing learning (Donovan et al., 2018). Therefore, task complexity may be desirable to thriving individuals who believe that job characteristics satisfy their needs of progress and growth. Conversely, low-complexity tasks provide little opportunity to grow and progress (Srikanth & Jomon, 2020), so if the task is simple, thriving individuals may be dissatisfied with their work.

Some indirect empirical evidence suggests that the positive effects of thriving may depend on the type of task being performed. Donovan et al. (2018) found that individuals holding a mastery goal (focus on gaining knowledge, enhancing their ability, and increasing learning) demonstrated higher levels of task enjoyment when asked to perform a more complex task as compared to a less complex task. Similarly, Steele-Johnson et al. (2000) indicated that participants with a learning goal orientation were more satisfied with their performance on a difficult task, which offered a greater opportunity to master the task at an early stage of skill acquisition. These results suggest that thriving workers may enjoy challenging tasks more than simple tasks.

Based on this, the following hypotheses are proposed:

Hypothesis 3: Task complexity moderates the relationship between thriving and N-S fit. As task complexity increases, the positive relationship between thriving and N-S fit strengthens.

Hypothesis 4: The indirect effect of thriving on task focus through N-S fit perceptions is conditional on increasing complexity.

Methods

Participants and procedure

The participants of our study were 170 product engineers and their line supervisors from a heavy industrial machinery company in China that has been ranked among the Fortune Global 500 companies. The research protocol was approved by the Biomedical Research Ethics Committee of our university (2021 – 413).

We first visited the management of the company, explained the purpose of our research and detailed the content of our study. After obtaining permission and support for data collection, the HR department provided us with the email addresses of the product engineers and their supervisors. We sent an initial recruiting email to 245 product engineers and 32 supervisors asking them to volunteer to participate in the survey. Unique codes for individuals and groups were used as identifiers to match the responses of employees and supervisors. All participants were assured that their information will be kept confidential, and that the data obtained will only be used for academic purposes.

A two-wave design was used to explore causal relationships among study variables. At time 1, product engineers were asked to recall and rate their thriving and task focus over the past week. Three weeks later (time 2), the second round of questionnaires was administered to engineers who completed the first wave, asking them to evaluate their task focus and N-S fit over the past month. At time 2, the supervisors were asked to rate the participants' task complexity.

The reasons for our design are as follows. First, in regard to the 1-week retrospective measurement of thriving, according to the continuum depicted by Youssef and Luthans (2007), we argue that, consistent with other positive psychology concepts (e.g., hope, optimism and resilience), thriving is a state-like psychological construct. Thriving has both a cognitive (learning) and an affective (vitality) component, and so may be somewhat stable and not change with each momentary situation in the manner of purely positive affect. Indeed, several studies have indicated that the learning dimension does not vary substantially within persons from day to day (e.g. Niseen et al., 2012; Rokitansky, 2018). George and Zhou (2002) showed that a 1-week time frame captured participants' mid-range mood states that, in terms of stability, fall between mood at the moment and traits. Second, in regard to the two-wave feature, there may be a delay between holding a certain cognition (e.g. learning and growth) and enacting an associated behavior (e.g. focusing on a task) (Wright and Staw, 1999), so a two-wave design leads to a better evaluation of causal effects. Finally, Wright and Staw (1999) argued that if the variation in performance between subjects is caused by positive state, then an overlap in time is necessary when measuring state and performance. Therefore, following the empirical practice of Tsai et al. (2007), we placed a 3-week gap between the measurement of thriving and task focus, while ensuring 1-week overlap between the two since participants were asked to reflect on their tsk focus and N-S fit over the last month.

Of course, the separation of predictors and dependent variables is also consistent with the now conventional guidelines proposed by Podsakoff and colleagues (Podsakoff et al., 2003) to proactively manage common method variance.

Excluding the data that was not matched across the two time points and participants with less than six months of job tenure, our final usable sample consisted of 170 engineers, a 69% return rate on our recruiting email. All 32 of the supervisors agreed to participate in the research. This represents a valid return rate of 69.3%. Of the participants, 112 were male and 58 were female, with an average age of 28.95 years, and an average job tenure of 5.31 years. The sample was highly educated with most of participants (164) holding a bachelor's degree or higher.

Measures

Thriving at Work. Ten items developed by Porath et al. (2012) were used to measure this construct, comprised of five items for learning, such as "I feel alive and vital", and five items for vitality, such as "I continue to learn more as time goes by". We followed the time frame adopted by George and Zhou (2002) as well as Tsai et al. (2007) who asked employees to rate how they had felt "during the past week" for each item.

Task Focus. Respondents rated their own task focus "over the past month" at both time 1 and time 2 with a three-item scale adapted from Rothbard (2001). An example item is "Over the past month, I focused a lot on my work".

N-S Fit. The N-S Fit scale of three items was developed by Edwards et al. (2006), an example being "there is a good fit between what my job offers me and what I am looking for at work." Since in our study we considered N-S fit to be a perception affected by an individual's thriving, it was necessary to provide a time frame for the participants. As a result, we asked respondents to assess retrospectively their general belief about the N-S fit "over the last month."

Task Complexity. Four items were used to measure task complexity with three items from Maynard and Hakel (1997, i.e., "I think his tasks are complex"; "I think his tasks required quite a bit of thought and problem solving"; "I think his tasks are challenging") and one item form Steele-Johnson et al. (2000, i.e., "I think his tasks are difficult to understand"). Although task complexity could be measured from both objective and subjective perspectives, Schwab and Cummings (1976) indicated that self-rating may confound the task performers' personal needs and preferences with the objective characteristics of tasks. Since thriving is also related to individual needs and preferences, serious endogenous problems may result if both task complexity and thriving are evaluated subjectively. As a result, in line with the Leplat's (1988) view of "supposed complexity", we asked the person prescribing the task (i.e., the supervisor) to evaluate the engineers' task complexity "over the past month." Also, task complex as evaluated by the supervisors can reduce common method variance. The modifications of task complexity measurement are as follows.

All of the above scales using five-point Likert scale (1 = strongly disagree to 5 = strongly agree).

Control variables

We controlled for the effects of gender, age, education, and tenure, considering their potential effects on effort, cognition, and performance (Brown & Peterson, 1993). Furthermore, due to the coherence and continuity of individual cognition and behavior, pre- and post-behavioral performance may be closely related in a month, so the task focus reported by participants may simply be a continuation of past behavior rather than the result of changes in psychology state or work environment. Based on this, task focus at time 1 was also controlled to eliminate spurious correlations.

Results

Reliability and validity

Harman's single-factor model approach was used to test for common method variance (CMV). As shown in Table 1, the variance explanation rate of the first factor was less than 40%, suggesting that CMV was not serious. Adding a latent variable did not significantly affect the model. The fit indices of the four-factor model were significantly better than the alternative models, with all indicators meeting and surpassing fit criteria ($\chi2/df = 1.20$ (≤ 3), FI = 0.975 (≥ 0.9), TLI = 0.970, RMSEA = 0.047 (≤ 0.08), and SRMR = 0.065 (≤ 0.08)), demonstrating good fit and discriminant validity for the four variables. In addition, the results of exploratory factor analysis showed that the factor loadings of all items were greater than 0.5 and the average variance extracted values (AVE) of all scales were greater than 0.5. The factor structure of the variables was good, and the variables had good convergent validity.

Correlation and descriptive statistics

The means, standard deviations, and correlations among variables are shown in Table 2. The results indicated that thriving was significantly and positively correlated with N-S fit perception and task focus at both T1 and T2. N-S fit was also significantly and positively correlated with TF both at T1 and T2. The significant relationship between the variables provided preliminary evidence for our hypotheses. In addition, TF at T1 andT2 were significantly correlated, indicating a strong consistency of behavior between the different times.

Table 1 Confirmatory factor analysis results ($N = 170$)

Model	χ^2	df	RMSEA	CFI	TLI	SRMR
Four-factor mode	189.538	158	0.047	0.975	0.970	0.065
Unmeasurable latent factor model	191.460	160	0.047	0.975	0.970	0.065
Three-factor mode	523.086	167	0.155	0.716	0.677	0.130
Two-factor mode	588.374	169	0.167	0.665	0.624	0.132
Single-factor mode	829.159	170	0.209	0.474	0.412	0.166

TAW: thriving at work, *N-S:* Needs-Supplies fit, *TF:* task focus, *TC:* task complexity

Single factor: TAW + N-S + TF + TC; Two factors: TAW + TC, TF + N-S; Three factors: TAW + TC, N-S, TF; Four factors: TAW, TC, TF, N-S

Table 2 Variable description statistics and correlation analysis ($N = 170$)

Variable	M	SD	1	2	3	4	5	6	7	8
1 Gender	1.64	0.48	NA							
2 Age	28.95	4.97	-0.14	NA						
3 Education	3.40	0.64	-0.02	0.32**	NA					
4 Tenure	5.31	4.47	-0.14	0.69***	0.15	NA				
5 TF 1 (T1)	3.00	0.88	-0.14	0.25*	0.04	0.60	NA			
6 TAW (T1)	3.56	0.68	-0.19	0.25*	0.08	0.18*	0.642***	NA		
7 N-S fit (T2)	3.41	0.79	-0.14	0.25**	0.22*	0.13	0.454***	0.48***	NA	
8 TF 2 (T2)	3.16	0.90	-0.17	0.18	0.291**	0.05	0.533***	0.52***	0.67***	NA
9 TC (T1)	2.87	0.92	-0.04	0.00	-0.13	-0.14	0.15	0.16	0.00	0.20

$N = 170$. Gender: Female = 1, male = 2. Education: 1 = high school or below, 2 = junior college, 3 = bachelor, 4 = Master or above

Alphas on the diagonal

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Table 3 Regression analysis ($N = 170$)

Variables	*N*-S Fit(T2)				Task Focus(T2)				
	M1-1	M1-2	M1-3	M1-4	M2-1	M2-2	M2-3	M2-4	M2-5
Gender	-0.11	-0.07	-0.12	-0.07	-0.21	-0.16	-0.14	-0.12	-0.20
Age	0.01	0.01	0.01	0.01	-0.01	-0.01	-0.02	-0.02	-0.02
Education	0.23*	0.21	0.23	0.24*	0.41**	0.39**	0.28*	0.28*	0.41**
Tenure	0.01	0.00	0.00	-0.01	0.00	-0.01	-0.00	-0.01	0.01
TF (T1)	0.38***	0.21	0.38***	0.19	0.53***	0.36**	0.32**	0.24*	0.515***
TAW(T1)		0.34*		0.40**		0.36*		0.18	
TC(T2)			-0.07	-0.14					0.11
Interaction				0.20**					
N-S Fit(T2)							0.57***	0.53***	
R^2	0.26	0.31	0.27	0.39	0.37	0.41	0.56	0.57	0.38
ΔR^2	0.21	0.25	0.21	0.32	0.33	0.37	0.52	0.53	0.34
F	5.53***	5.32*	4.69***	5.96***	9.32***	5.56*	16.28***	27.13	8.05***

Hypotheses testing

We conducted hierarchical regression analyses using SPSS 27. Table 3 presents the results for Hypotheses 1 through 3. Model 2–2 confirmed Hypothesis 1, that thriving at T1 was positively related to task focus at T2 ($p = 0.021$), explaining 37% of the variance in task focus T2.

For Hypothesis 2, in Model 2–4 the regression coefficient of the mediating variable N-S fit was significant ($p = 0.000$), but the regression coefficient of the independent variable thriving was no longer significant ($p = 0.191$). This suggests that N-S fit played a fully mediating role between thriving and task focus, supporting Hypothesis 2.

For test of Hypothesis 3, we first centered thriving and task complexity to avoid multicollinearity among explanatory variables. Adopting the causal steps approach proposed by Muller et al. (2005), we found that the interaction between thriving and task focus was positively associated with N-S fit in Model 1–4 ($p = 0.004$), supporting Hypothesis 3. Simple slope tests revealed that the link between thriving and N-S fit was positive when task complexity was high ($p = 0.000$) but was not significant when task complexity was low ($p = 0.514$) (see Fig. 2).

Finally, a bootstrapping test for the moderated mediation model was conducted following the methods proposed by Preacher et al. (2007), with a sample size of 5000 selected, 95% confidence interval (CI), and one standard deviation added/subtracted as the low and high conditional values. Conditional effects of thriving on N-S fit were significant at high task complexity ($\beta = 0.685$, $p = 0.000$, 95% CI [0.326, 1.045]) and mid-level complexity ($\beta = 0.396$, $p = 0.006$, 95% CI [0.115, 0.678]), but not at low task complexity ($\beta = 0.107$, $p = 0.514$, 95% CI [-0.219, 0.433]). The bootstrapping results presented in Table 4 indicate that the indirect relationship between thriving and task focus via perceived N–S fit contained zero when task complexity was low, but not when it was high. These results are in line with Hypothesis 4, that the indirect positive effect would be stronger in the context of greater task complexity.

Discussion

Besides offering one of the first studies to integrate P-E fit theory with the socially embedded model of thriving, a goal of this study was to better understand the relationship

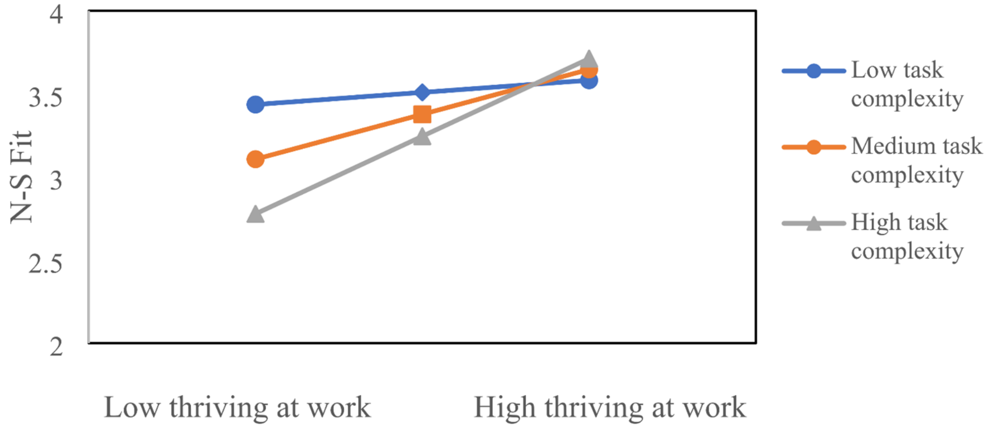

Fig. 2 Moderating effect of task complexity on the relationship between TAW and N-S fit

Table 4 Direct and mediating effect

TC	Mediating effect TAW→N-S Fit→ TF				Direct effect TAW→TF			
	Effect	SE	Bootstrap's 95%CI		Effect	SE	Bootstrap's 95%CI	
			LLCI	ULCI			LLCI	ULCI
-0.917	0.057	0.096	-0.127	0.259	0.101	0.154	-0.205	0.407
0.000	0.211	0.086	0.074	0.426	0.169	0.139	-0.108	0.446
0.917	0.364	0.114	0.183	0.651	0.237	0.185	-0.131	0.605

between thriving and task focus. We proposed a model with N-S fit as the mediator and task complexity as the moderator, to explain how and when job complexity translates into task focus. Perhaps most importantly, placing thriving as the independent variable rather than the outcome of fit and task focus, our findings provide insight into a possible iterative, sustainable mechanism for thriving at work.

The results supported our hypotheses, indicating that thriving positively predicts later task focus, even controlling for earlier task focus (H1), and that this relationship is mediated by N-S fit (H2). This core of our model suggests a possible iterative mechanism for thriving at work that may be self-sustaining. Further, the indirect effect of thriving on task focus through N-S fit was moderated by task complexity. To be specific, the positive mediation held at higher levels of task complexity but was not significant when the task was not complex.

An intriguing observation about this interaction is that high complexity is detrimental when thriving is low, negatively impacting on N-S fit. However, another interpretation is that this negative impact lessens as complexity also lessens, so that low thriving individuals in low complex tasks have as much task focus as high thriving individuals in highly complex tasks. Considering the iterative model of thriving, N-S fit, and task focus, this could suggest that organizations and managers should consider tailoring task complexity to individual's initial levels of thriving, progressively increasing complexity as N-S fit feeds back to promote thriving through task focus, essentially developing a resource the employee can draw upon to cope with complexity. Of course, there are likely be boundary conditions on this reciprocal engine of thriving and complexity, but other interventions could be applied to make thriving more viable and reasonable to employees as they adjust. An iterative, progressive approach may be effective in a context in which employees are already overwhelmed by their work.

Theoretical implications

First, our study substantiates the positive effect of thriving on agentic behavior, which we operationalized as task focus, providing empirical support for this important feature of the SEMT. To the best of our knowledge, only one study has to date empirically tested the thriving-task focus relationship (Niessen et al., 2012). Our work addresses this gap while also responding to the recent calls for more studies that examine the causal path in the SEMT (Goh et al., 2022; Kleine et al., 2019). Although our result is somewhat at odds with Niessen et al. (2012), who found no significant relationships between thriving and task focus, it is important to note that there are several crucial differences that may explain this discrepancy. Niessen et al. focused on short-term fluctuations in task focus, assessing it the day immediately following the measurement of thriving. Our study took a one-month retrospective assessment of task focus. As pointed out by Sonnentag et al. (2012), even employees who remain highly engaged in their work over long periods are likely to experience ups and downs in engagement during a given period. This may mean that while momentary effects of thriving on focus will fluctuate and be influenced by other events, thriving can still be a significant predictor of overall task focus. Therefore, to more accurately describe the relationship between thriving and task focus, future research may benefit from adopting different time frames (hour, day, week, month and even year) for empirical tests. In addition, the social services work environment that formed the context of Nissen et al.'s study is not the same as that for the engineers in our study. Social services workers need to have extensive contact and interaction with different unemployed individuals (Niessen et al., 2012), and their task focus may change in a short time due to specific work events and clients. For engineers, their work resources, customers, and environment are relatively stable, so the predictive effect of thriving at work may be stronger. This suggests that particular models and findings for the thriving and task focus relationship may not be generalizable to all types of work, and that task or work context is an important moderator. More research is needed to clarify context as a boundary condition.

Second, this study expands the scope of the SEMT by identifying N-S fit as an underlying mechanism of the relationship between thriving and task focus. Some of the literature has asserted that individuals may use their thriving to gauge whether they are on a positive development path, leading them to sustain or change their agentic behavior (e.g., Spreitzer et al., 2005). However, this proposition is often assumed but rarely rigorously modeled or empirically examined (Goh et al., 2022). Our study demonstrates that the fit perspective can provide an explanatory framework. Our findings are generally consistent with the basic tenet of the affect-based model of P-E fit proposed by Yu (2009), whereby work feelings predict work behavior through shaping a sense of fit. This path is supported by Cho et al. (2009) who found that employees who experience thriving at work are likely to perceive their work environment as supportive of their self-development and goal pursuit, thereby enhancing their willingness to stay in that environment. We suggest that future research could more broadly explore the pipeline of fit perception in the SEMT.

Third, by showing that task complexity is a critical contingency in the thriving – N-S fit relationship, our study enriches the context-embedded research on the impact of thriving at work. We found that the more complex the task, the stronger the relationship between thriving and N-S fit. These findings echo previous studies showing that thriving employees may seek and enjoy assignments that enable them to acquire additional knowledge and skills (e.g., Porath et al., 2012; Chew, 2017). On a broader level, our findings are consistent with the hypothesis of the job characteristics model proposed by Hackman and Oldman (1976) that only employees with high growth needs will have positive consequences such as high job satisfaction in challenging jobs.

Finally, our model integrates the thriving with fit literature, providing directions for fusion research in these two fields. Our findings enrich the nomological connections of fit by examining both antecedents and consequences of fit in the same model. We found that employees who thrive at work reported higher N-S fit, suggesting that thriving may be a strong predictor of fit perception.

Practical implications

A significant applied contribution of this study draws on the demonstration that complexity of task can influence the relationship between thriving and task focus. This finding opens up the possibility for organizations to actively facilitate individual's thriving by matching them to the characteristics of different work tasks. For example, it may be that vitality and learning would be appropriate when employees are working on complex and challenging tasks, for example the design and development of new products. Further, the insight generated by our study could help managers assign the right people to the right work tasks at the right time. Employees who are energetic and enjoy learning could be matched with complex tasks, while employees experiencing challenges or burnout with less complex tasks. Such a matching approach could harness the iterative mechanism by which thriving can be promoted and sustained. Nevertheless, in the case of matching employees with less complex tasks, this strategy should be complemented by other thriving promotion initiatives and communication that the task assignment is not a demotion or constraint on the employee's growth and progress, but rather meant to assist them recover their energies (Porath et al., 2022). These types of organizational practices would impact the cognition and motivation related to the employees' behavior and performance (e.g. Donovan et al., 2018).

A second implication of this study is that, more generally, organizations should promote and maintain employees' thriving, since thriving is significantly correlated with perceived fit and task focus. A recent metanalysis of thriving indicated the key antecedents of thriving at work include supportive coworker behavior, supportive leadership behavior and perceived organizational support (Kleine et al., 2019). For example, For the engineers in our study, managers can provide the autonomous at the workplace, adopt correct interactive procedures and delightful interactive behavior, and communicating that their employee's work is meaningful, thus contributing to feelings of vitality and learning (e.g., Farid et al., 2023). In addition to organizations providing a socially embedded context for thriving, workers also sustain their own thriving by the way of sleep, breaks, helping others and so on (Porath et al., 2022).

Limitations and future research

This study also had some notable limitations that call for additional investigation in the future. First, we included only one type of agentic behavior, task focus, in our study. While task focus is an important variable in the literature on thriving and fit, future research could explore the effects of TAW across a broader spectrum of behavioral performance, which may lead to different predictions.

Second, we examined only a single job characteristic, task complexity, because it is a central variable in N-S fit perceptions research and is also considered a general job characteristic that includes a number of specific factors (e.g., diversity, workload, and skill level) (Wood, 1986). In general, an individual's perception of his or her job's needs-supply fit involves many aspects, with different characteristics producing different modes of moderation or reinforcement (Edwards et al., 2006). We encourage researchers to investigate the differential moderating effects of particular job characteristics, such as job variety and workload. In addition, person-environment fit is a generalized term, but our study focuses only on the dimension of N-S fit. The perception of N-S fit derives more from context and experience than individual traits or characteristics and is also closely related to work and career outcomes (Edwards et al., 2006). Therefore, the theoretical link between N-S fit and thriving may be stronger than demands-abilities or other forms of fit (Kristof-Brown et al., 2005). Nevertheless, there may be valuable contributions in the exploration and integration of thriving with other forms of fit on a multidimensional basis, such as demand-ability fit, individual-organization fit, and individual-supervisor fit.

Third, in terms of research design, we measured variables at different time points, for reasons associated with measurement of the key variables (George & Zhou, 2002; Tsai et al., 2007; Wright & Staw, 1999) and also to reduce common method variance (Podsakoff et al., 2003). Some of the relationships in the reported statistical models, however, might have been overstated because most of the variables were examined using self-rated data. We suggest that multiple sources of data collection could be utilized in future studies. Like other scholars, we also encourage the application of laboratory or diary methods in future research in order to more precisely track changes and effects of thriving.

A final point worth discussing is the sample used in this study. Our study was based on a sample of engineers in the manufacturing industry, which might limit the generalizability of the results. In order to further test Spreitzer et al. (2005)'s socially embedded model of thriving, a broader study of employees in other occupational groups and industries is needed. In addition, the data was collected in China, so the cross-cultural universality of the results may be an issue. Future studies could use samples from Western societies to test our model.

Conclusion

Thriving and productive employees are valuable to managers and organizations. In this study, we theorized and tested a moderated mediation mechanism by which thriving might be self-sustainable. We incorporated the person-environment perspective into the socially embedded model of thriving to explore the proposition that thriving at work predicts task focus, which is usually considered an antecedent rather than an outcome of thriving. The mediator of needs-supplies fit was key to understanding the self-sustainability proposition. Our findings demonstrated that employees who feel that their work environment satisfies their needs will be more likely to engage in task focus, an antecedent of thriving at work. Critically, needs-supplies fit can be maintained and promoted by a strategy of matching an employee's current state of thriving with the level of task complexity. Our study extends the understanding and knowledge about thriving and fit, while providing managers with a person-task matching strategy that can be an iterative mechanism that promotes and sustains thriving at work.

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Data availability The datasets generated during and/or analyzed during the current study are available in the "Figshare" repository, https://doi.org/10.6084/m9.figshare.25422637.

Declarations

Ethnical approval The authors have no relevant financial or non-financial interests to disclose. The study was approved by the Biomedical Research Ethics Committee of Hunan Normal university (2021 – 413) and was performed in accordance with the 1964 Helsinki Declaration.

Informed consent Informed consent was obtained from all individual participants included in the study.

Conflict of interest There is no potential conflict of interest with respect to the research, authorship, and publication of this article.

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